Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

1. **Recommendations**:

   Based on the analysis of shipping performance, the following recommendations are proposed to improve logistics efficiency:
   A. Route Optimization:
   Routes with consistently high lead times should be re-evaluated and optimized. This may involve redesigning delivery paths, adjusting distribution strategies, or redistributing load across alternative routes.
   B. Focus on High-Delay Regions:
   States identified as geographic bottlenecks require targeted intervention. Improvements in infrastructure, warehouse placement, or delivery scheduling can significantly reduce delays in these regions.
   C. Shipping Mode Optimization:
   Shipping methods with lower lead times should be prioritized for high-priority deliveries. Organizations should evaluate the trade-off between cost and delivery speed to select the most efficient shipping modes.
   D. Reduce Variability in Delivery Time:
   High variability indicates inconsistency in operations. Standardizing logistics processes and improving coordination between factories and distribution centres can enhance reliability.
   E. Implement Real-Time Monitoring Systems:
   The organization should adopt real-time tracking and monitoring systems to continuously evaluate shipping performance and detect delays early.
   F. Data-Driven Decision Making
   The use of dashboards and analytics tools should be integrated into daily operations to support proactive logistics management rather than reactive problem-solving.
   These recommendations aim to improve delivery efficiency, reduce operational delays, and enhance overall supply chain performance.

Executive Summary

Efficient logistics systems are essential for ensuring timely delivery, reducing operational costs, and improving overall supply chain performance. This study analyses factory-to-customer shipping routes within a distribution network to identify inefficiencies, delays, and geographic bottlenecks.

Using a dataset of over 10,000 shipping records, key performance metrics such as average delivery time, variability, and delay rates were evaluated. The analysis revealed significant disparities in route performance, with certain routes consistently outperforming others, while some exhibit frequent delays.

Geographic analysis identified specific states with high delivery times, indicating bottlenecks that require targeted intervention. Additionally, the comparison of shipping methods highlighted differences in performance, suggesting opportunities for optimization.

An interactive dashboard was developed to enable real-time monitoring and analysis of shipping performance. This tool allows stakeholders to explore data dynamically, apply filters, and identify inefficiencies quickly.

The findings of this study emphasize the need for a transition from reactive to data-driven logistics management. By implementing route optimization strategies, improving infrastructure in high-delay regions, and leveraging analytics tools, organizations can significantly enhance delivery efficiency and operational effectiveness.

This project provides a scalable framework that can be extended to larger logistics systems, including government and public sector supply chains, to improve service delivery and resource utilization.